

# HAT-P-36b

R-1909

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-36\_250608 · created 2026-08-05T23:29:51+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460834.7382 +/- 0.0047 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.157 +/- 0.036                 |
| Transit depth (Rp/Rs)^2                           | 2.46 +/- 1.14 [%]               |
| Orbital Inclination (inc)                         | 81.3 +/- 2.44                   |
| Airmass coefficient 1 (a1)                        | 1.0 +/- 0.02                    |
| Airmass coefficient 2 (a2)                        | 0.003 +/- 0.018                 |
| Scatter in the residuals of the lightcurve fit is | 2.01 %                          |
| Best Comparison Star                              | #3 - [150, 404]                 |
| Optimal Aperture                                  | 3.11                            |
| Optimal Annulus                                   | 16.53                           |
| Transit Duration (day)                            | 0.0353 +/- 0.0088               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.157 $\pm$ 0.036 vs 0.126 (deviation 0.86 $\sigma$ ) |
| Duration fit vs published            | 0.0353 d vs 0.0925 d (deviation 6.5 $\sigma$ )        |
| Mid-time O-C                         | -15.3 min                                             |
| Depth SNR                            | 4.4                                                   |
| Residual scatter                     | 2.01 %                                                |

### Light curve

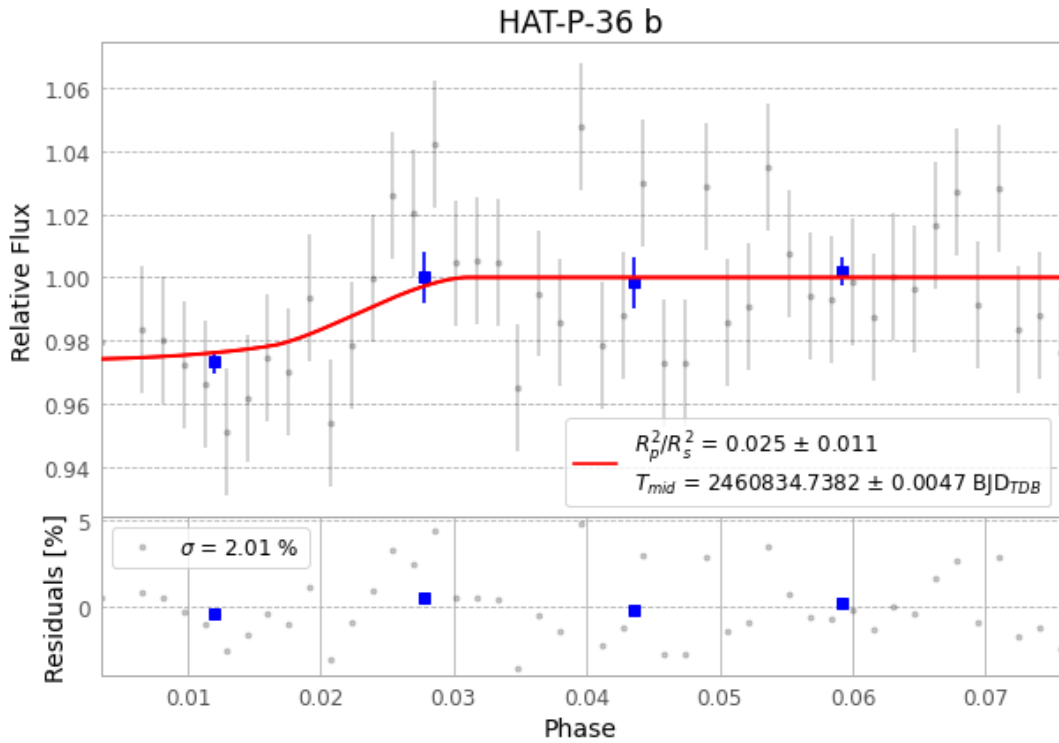


Figure 1 — FinalLightCurve\_HAT-P-36 b\_2025-06-07.png (SHA-256 ee345f8a7f4bc5a3...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 257], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d7e9c41eb3452c8d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

47 FITS frames (31 MB), HATP-36250608054816.FITS → HATP-36250608080616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250608.log (8a4918e7b705...), FinalLightCurve\_HAT-P-36 b\_2025-06-07.png (ee345f8a7f4b...), FinalLightCurve\_HAT-P-36 b\_2025-06-07.csv (72bf8660eb15...)

## Compute resources

Wall time 147.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:29Z.